

QUALITY ASSURANCE PLAN

Medkart Savings Journey

MVP test strategy, defect workflow and pilot release controls

STATUS	Draft for prototype validation and implementation feasibility review
RELATED ARTIFACTS	PRD v0.1 • Feature Specification v0.1 • QA Tracker workbook
MVP SCOPE	Linked purchase history, digital invoice, explainable savings and correction handling
EXECUTION STATE	40 test cases designed; execution status is Not Run
EVIDENCE STATE	Eight sample defects demonstrate reporting structure; none are production claims
VERSION	0.1 • 19 July 2026
DISCLOSURE	Independent candidate exercise; no access to Medkart production systems

Quality objective Prove that store-purchase records are linked, explained and corrected without exposing another customer’s data, misstating savings or interrupting store checkout.

Evidence rule A designed test is not a passed test. A sample bug is not an observed defect. Execution status changes only after authorized testing with reproducible evidence.

1. Scope and quality risks

1.1 In scope

- Transaction ingestion, eligibility, idempotency and delayed-event recovery.
- Unique account matching, fail-closed ambiguous/missing matches and authorization.
- Purchase history, detail, digital invoice and transaction lifecycle states.
- Approved savings calculation, explanation, unavailable states and rounding.
- Issue reporting, immediate protection for “not mine,” case status and support handoff.
- Accessibility, privacy-safe analytics, operational readiness and controlled pilot release.

1.2 Out of scope

- Clinical accuracy, prescription validation or medicine-consumption inference.
- Family assignment, refill reminder consent integration and reorder flows.
- Penetration testing of any live environment without written authorization.
- Validation of Medkart’s current internal retail software or production architecture.

1.3 Risk-based priorities

Risk	Why it matters	Primary controls	Gate
Cross-account exposure	Purchase records may contain sensitive transaction information.	Fail closed; endpoint authorization; cache isolation; audit.	P0
Incorrect identity link	Shared/recycled numbers or normalization can mis-link records.	Unique verified match; ambiguous state; protected claim.	P0
Misstated savings	Unapproved/missing comparator can damage trust.	Approved method/version; suppression; reconciliation.	P0
Duplicate or stale records	Retries/refunds can corrupt history and totals.	Idempotency; ordering; lifecycle reconciliation.	P0
Checkout dependency	Customer billing must continue during linking failure.	Asynchronous event; queue; recovery monitoring.	P0
Support dead end	A wrong record must be correctable quickly.	Case ID; ownership; status; escalation SOP.	P1

2. Test strategy

Layer	Focus	Examples	Owner
Component	Calculation and state logic	Eligibility, normalization, savings, lifecycle reducer	Engineering
API / contract	Authorization and event behavior	Idempotency, replays, error taxonomy, audit	Engineering + QA
Integration	Retail-event to app journey	Matched, delayed, refund, outage recovery	QA + Engineering
End-to-end	Customer-visible core flow	History → detail → invoice/savings → issue	QA
Security/privacy	Data isolation and minimization	Direct URL, shared device, analytics, brute force	Security/Privacy
Accessibility	Operability and comprehension	Keyboard, screen reader, scaling, status semantics	QA + Design
Operations	Support and release readiness	Case queue, SOP, alerts, rollback drill	Product Ops

2.1 Test data principles

- Use synthetic accounts, stores, invoices and item labels only.
- Create explicit datasets for unique, missing, ambiguous and recycled-phone conditions.
- Keep a deterministic event ID for replay and out-of-order lifecycle testing.
- Do not copy customer phone numbers, medicine names, prescriptions or support notes into test evidence.
- Version the approved savings comparator and record expected calculations independently.

2.2 Environments and observability

Need	Minimum evidence
Authorized staging or isolated test environment	Environment name, build/version, feature flag and tester.
Event visibility	Event ID, processing state, retry count and reconciliation outcome.
Authorization evidence	Request outcome and redacted audit event; never include another user's detail.
Analytics validation	Approved schema, one-event behavior and redacted payload inspection.
Support validation	Case ID, routing, status transition and SLA/owner confirmation.

3. Core scenario set

The executable workbook contains 40 detailed test cases. The following set is the interview-level walkthrough.

ID	Scenario	Expected outcome	Priority
TC-001	Eligible matched purchase	One linked record with correct store/date/amount/status.	P0
TC-002	Duplicate event replay	One record; savings counted once.	P0
TC-005	Ambiguous account match	No transaction detail exposed; protected state.	P0
TC-013	Missing comparator	Savings amount suppressed with honest explanation.	P0
TC-015	Full return	Returned state; amounts and savings reconciled.	P0
TC-018	“Not mine” issue	Detail protected immediately; case created.	P0
TC-021	Direct URL from wrong account	Denied, no sensitive response, attempt logged.	P0
TC-027	Linking outage	Checkout completes; event reconciles later.	P0
TC-032	Sensitive issue note	Free text excluded from general analytics.	P0
TC-040	No reminder consent	Purchase does not schedule a reminder.	P0

3.1 Pass, fail and blocked

Status	Definition
Passed	Observed result matches expected result and required evidence is attached.
Failed	Observed result differs; link a reproducible defect.
Blocked	Execution cannot proceed because environment, dependency or data is unavailable.
Not Run	Designed but not executed. This is the workbook default.
Not Applicable	Reviewed and excluded with a recorded reason/approver.

4. Defect management

4.1 Severity model

Severity	Definition	Response
Critical	Cross-account exposure, systemic authorization failure or severe privacy/security harm.	Immediate no-go; restrict access; incident path.
High	Core data integrity, lifecycle, savings or correction failure with material user impact.	Release blocker until fixed or formally risk-accepted.
Medium	Recoverable functionality, accessibility or UX failure with a workaround.	Resolve before pilot unless explicitly approved.
Low	Minor presentation or clarity issue without material risk.	Prioritize by value and capacity.

4.2 Required bug report anatomy

- One specific summary: condition + behavior + impact.
- Environment/build, test ID and reproducible steps.
- Expected result, observed result and redacted evidence.
- Severity based on impact, not emotion; owner and target milestone.
- Privacy-safe data: no customer contact details, medicine names or clinical notes.
- Retest result, regression scope and closure evidence.

Sample-defect disclosure The workbook includes eight intentionally designed sample scenarios to demonstrate reporting quality. Their Evidence Type is “Sample scenario” and Status is “Draft.” They must not be presented as bugs found in Medkart.

4.3 Workflow

State	Exit condition
Draft	Reproduced in an authorized environment and triaged.
Open	Owner and severity accepted.
In Progress	Fix/build reference available.
Ready for Retest	Deployable build and test notes provided.
Closed	Original case and agreed regression set pass.
Deferred	Reason, risk owner and review date recorded.

5. Entry, exit and release decision

5.1 Test entry criteria

- PRD acceptance criteria and MVP scope are stable enough to test.
- Authorized environment, synthetic test data and required integrations are available.
- Savings comparator/method and state mappings are approved for the build.
- Logging is sufficient to diagnose without collecting unnecessary sensitive data.
- Known limitations and owners are recorded.

5.2 Pilot exit criteria

- All P0 cases pass; no P0 case is Not Run, Failed or Blocked.
- No open Critical or High privacy/security defect.
- No known duplicate counting, cross-account exposure or checkout-blocking behavior.
- Savings explanation and unavailable states are approved and reconciled.
- Support SOP, correction route, monitoring, feature flag and rollback are ready.
- Accessibility and analytics checks pass for the pilot scope.
- Go/no-go decision and risk acceptances are recorded by accountable owners.

5.3 Automatic no-go triggers

Trigger	Action
Any cross-account exposure or authorization bypass	Disable/restrict feature; security/privacy incident path.
Ambiguous identity reveals transaction metadata	Fail closed and stop rollout.
Savings is displayed without an approved comparator	Suppress module; stop rollout until corrected.
Linking failure interrupts checkout	Disable dependency; restore checkout independence.
Correction path cannot protect a “not mine” record	Stop rollout and activate support fallback.

Decision rule The workbook dashboard is a review signal, not an automated release authority. Product, Engineering, QA, Security/Privacy and Operations remain accountable for go/no-go.

6. Pilot, monitoring and rollback

6.1 Controlled pilot

- Use a feature flag and small approved cohort of stores/accounts.
- Confirm staff briefing, support route and escalation contacts before enabling.
- Monitor match outcomes, duplicates, reconciliation delays, issue categories and invoice availability.
- Review early cases daily; expand only after stable results and explicit approval.

6.2 Suggested operational indicators

Indicator	Purpose	Guardrail
Eligible event → linked record rate	Detect ingestion/match breakage.	Segment safely; never expose unmatched detail.
Ambiguous/missing match rate	Measure identity friction.	No automatic linking when uncertain.
Duplicate record rate	Detect idempotency failure.	Target zero; investigate immediately.
Reconciliation delay	Detect queue or lifecycle lag.	Alert on defined threshold.
“Not mine” issue rate	Detect possible mis-linking.	Privacy review and rapid protection.
Savings unavailable/error rate	Detect comparator/data gaps.	Suppress rather than estimate.
Support resolution time	Check operational readiness.	Do not promise an SLA until approved.

6.3 Rollback readiness

- Named decision owner and technical operator.
- Documented triggers, feature-disable steps and confirmation check.
- Checkout remains unaffected when the app feature is disabled.
- Queued events and linked records have an approved recovery/reconciliation plan.
- Customer/support communication copy is ready for material incidents.

7. Roles, cadence and evidence

Role	Accountability
Product Manager	Scope, acceptance criteria, savings meaning and go/no-go decision.
Engineering	Implementation quality, observability, fixes, feature flag and rollback.
QA Owner	Test design, execution evidence, defect lifecycle and release summary.
Security/Privacy	Identity, authorization, data minimization and incident response.
Retail Operations	Store impact, pilot readiness and checkout continuity.
Support	Correction workflow, case ownership, status language and escalation.
Analytics/Data	Approved events, comparator lineage and monitoring integrity.
Design/Accessibility	Comprehension, error states and accessible interaction.

7.1 Recommended cadence

- Daily during execution: failures, blockers, privacy risks and environment health.
- Before pilot: evidence review against the 17-item release checklist.
- During pilot: daily metric/issue review with a named decision log.
- After pilot: findings, unresolved risks, customer/store feedback and scale recommendation.

7.2 Evidence checklist

Artifact	Minimum content
Test run	Case ID, environment/build, tester, timestamp, observed result and evidence link.
Defect	Reproduction, expected/observed, severity, owner, target and redacted evidence.
Release control	Owner, completion state and proof/decision link.
Go/no-go record	Decision, attendees, open risks, acceptances and rollback owner.
Pilot summary	Scope, outcomes, metrics, issues, feedback and recommendation.

8. Interview walkthrough

- Open QA Dashboard and explain why it says execution pending.
- Show TC-005 to demonstrate fail-closed privacy thinking.
- Show BUG-001 and point to “Sample scenario / Draft” disclosure.
- Open Coverage Matrix to prove traceability to PRD requirements.
- Close with Release Checklist and the human go/no-go rule.

Suggested close “I have not tested Medkart production. This package shows how I would turn requirements into a safe, traceable QA and release process, then replace assumptions with authorized evidence.”